Automatic Toll Gate System

MCA Semester III

MCA-35 INTERNET OF THINGS

PROJECT REPORT

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF

MASTER OF COMPUTER APPLICATIONS

(S.K PATEL INSTITUTE OF MANAGEMENT AND COMPUTE STUDIES)

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Step 1. PURPOSE & REQUIREMENTS SPECIFICATION

1.1 Purpose

A sun-following solar tracker system that allows: -

1.1.1 Automatic Vehicle Detection.:

- The ultrasonic sensor detects when a vehicle is nearby and triggers the system to open the gate automatically.

1.1.2. Hands-Free Operation.:

- The system reduces human involvement, allowing vehicles to pass seamlessly without requiring the driver to press buttons or ask for access.

1.1.3. Smart, Cost-Effective Solution.:

- By using inexpensive components like Arduino, ultrasonic sensors, and servo motors, the system provides an affordable and efficient automation solution suitable for small to medium setups.

1.1.4 Enhanced Safety and Control.:

- The red and green LEDs provide clear signals to the driver—red means stop, green means it's safe to go—thus improving traffic management and avoiding accidents.

1.2 Requirements Specification

1.2.1. Behavior:

1). Initial State:

- The barrier gate is closed.
- The red LED is ON, indicating vehicles must stop.
- The green LED is OFF.

2). When a Vehicle Approaches:

- The system opens the barrier by moving the servo motor to 90°.
- The red LED turns OFF.
- The green LED turns ON, indicating the vehicle can pass.
- The system continuously monitors the distance to ensure the vehicle passes through safely.

3). When the Vehicle Passes:

- The servo motor moves back to 0°, closing the barrier.
- The red LED turns ON, indicating vehicles must stop.
- The green LED turns OFF.

1.2.2. System Management Requirement:

- The system should continuously monitor the distance using the ultrasonic sensor.
- It must control the servo motor for opening and closing the barrier.
- It should manage the red and green LEDs to indicate the gate's status.
- The system should process sensor data and control outputs in real time without manual intervention.

1.2.3. Data Collection Requirement:

- The system should collect distance signals from the ultrasonic sensor to detect the presence of vehicles approaching the gate.

1.2.4. Data Analysis Requirement:

- The Arduino should analyze the ultrasonic sensor readings to decide when to open or close the barrier, ensuring vehicles can pass safely.

1.2.5. System Management Requirement:

- The system must manage the servo motor adjustments for opening and closing the gate based on vehicle detection.
- The system should control the red and green LEDs to display the correct status.
- The system must continuously monitor the sensor's data and update the outputs without delay.

1.2.6. User Interface Requirement:

- The system should provide feedback through LEDs to inform users about the gate's status:
    - Red LED → Stop
    - Green LED → Go

1.2.7. Data Privacy & Security Requirement:

- The system uses only local hardware such as the ultrasonic sensor, servo motor, and LEDs.
- Since it is not connected to external networks, it minimizes the chances of unauthorized access or misuse.

Step 2. PROCESS SPECIFICATION

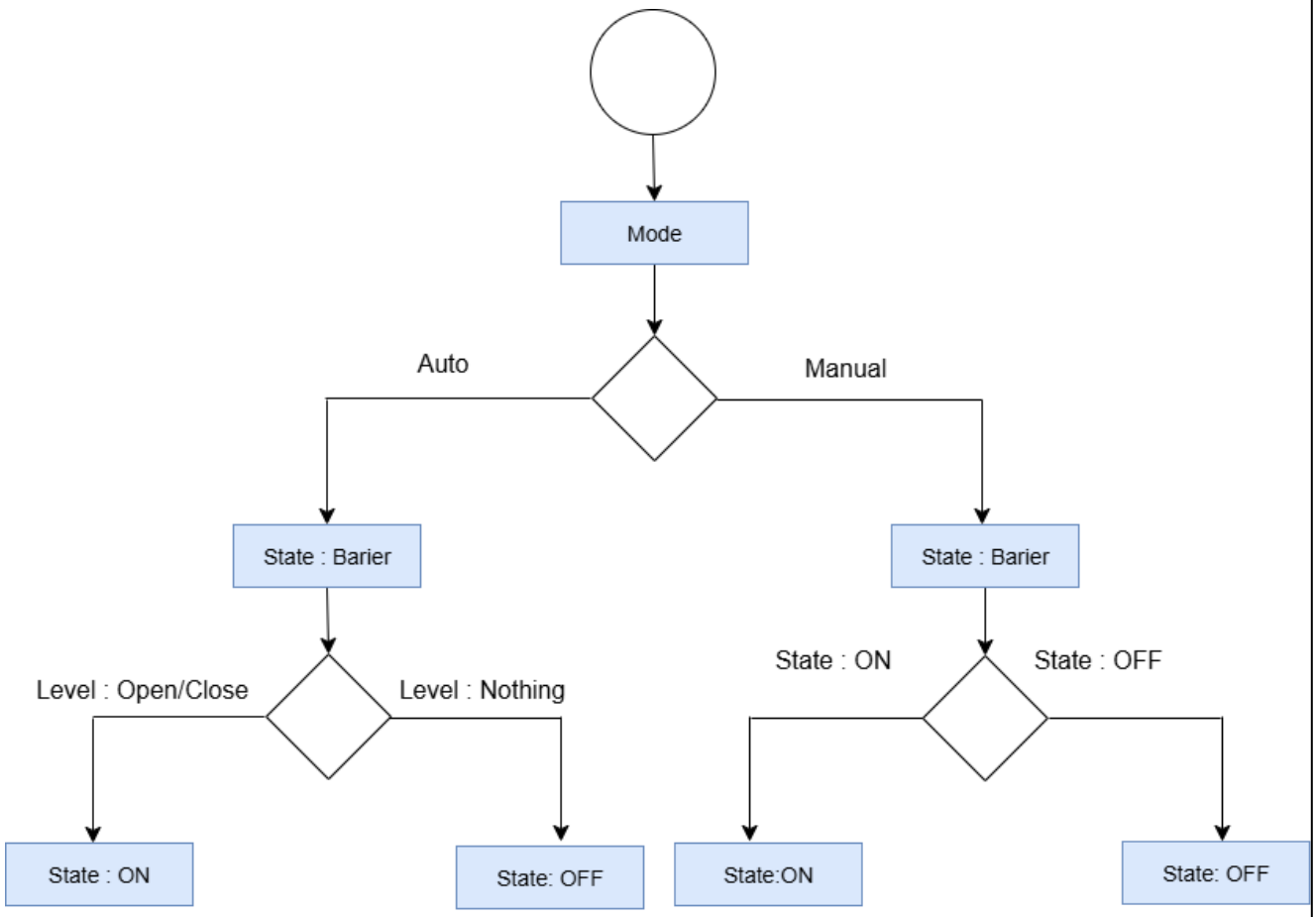


Fig. 2

Step 3. DOMAIN MODEL SPECIFICATION

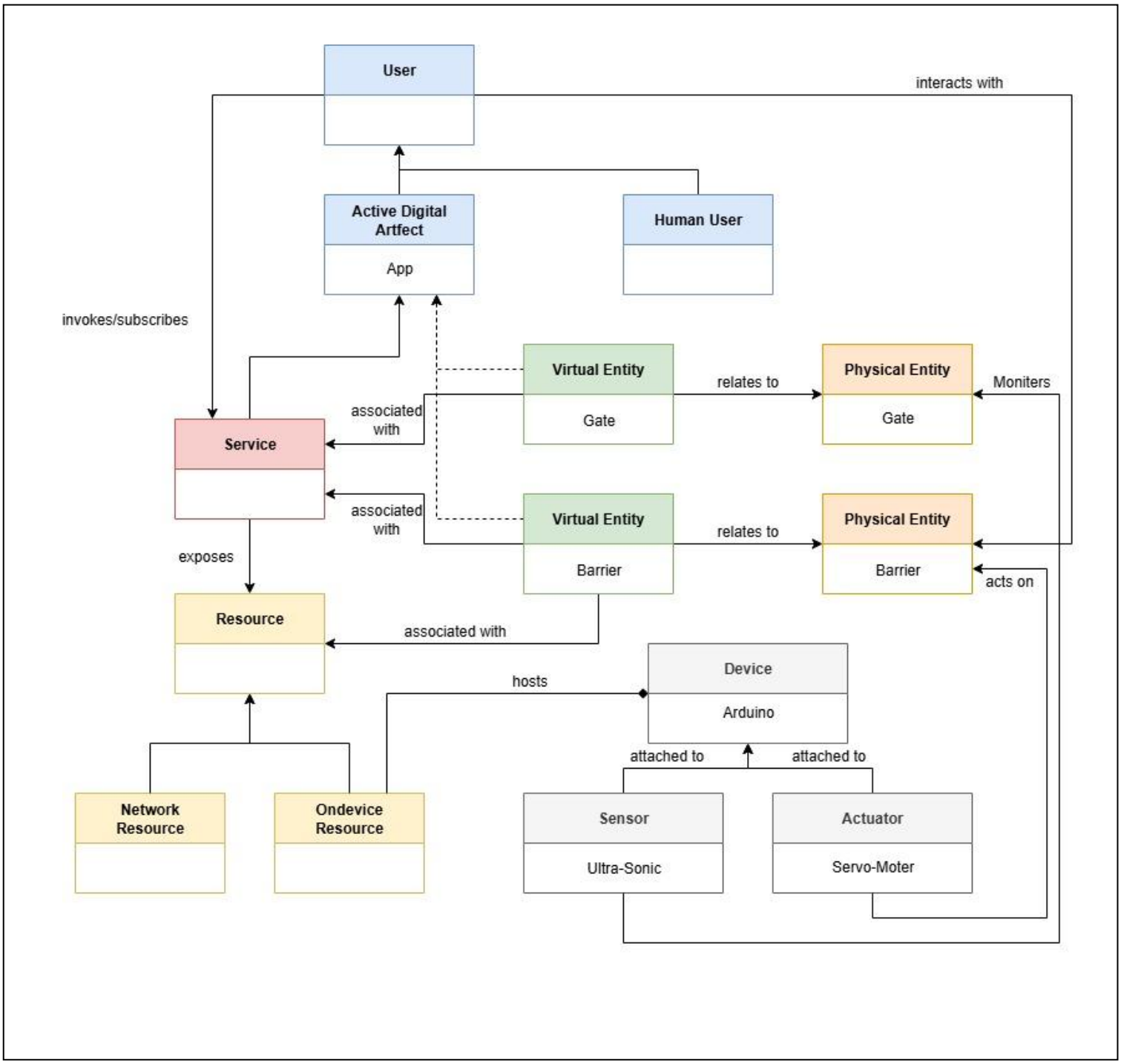


Fig. 3

STEP 3 <u>EXPLANATION</u>

- This diagram shows the structure and working of your **Automatic Toll Gate System** using IoT technology. It explains how users, devices, sensors, actuators, and services are connected and interact to control the toll gate automatically.
- User:
    - The **user** can be a vehicle driver or a toll system operator.
    - The user can interact with the system through an app or interface.
    - The user can monitor the toll gate's status and control it when necessary.
- Service:
    - The **service** is the core logic of the system.
    - It manages data from sensors and controls actuators.
    - The service is connected with virtual entities like the gate and barrier.
    - It exposes resources to the user for monitoring and control.
- Virtual Entity vs Physical Entity:
    - Virtual Entity (Gate/Barrier**)**: A digital representation of the gate and barrier.
    - Physical Entity **(**Gate/Barrier**)**: The real hardware installed at the toll gate.
    - The service uses virtual models to control and monitor the actual physical devices.
    - Sensors monitor the real gate's condition, while actuators act on it to open or close.

- Resource:

    - **Resource** represents the information or functions available in the system.

    - It can be a network resource (communication protocols) or on-device resource (sensor data).

    - Resources are associated with the gate and barrier, helping users access data or control the devices.

- Device – Arduino:

    - The Arduino is the central controller of the system.
    - It processes data from sensors and sends commands to actuators like the servo motor.
    - It executes the system's program to operate the toll gate efficiently.

- Sensor and Actuator:

    - Ultra-Sonic Sensor: Detects vehicles approaching the toll gate.
    - Servo Motor: Controls the barrier's movement by opening or closing it.

- Data Flow and Interaction:

    - The user interacts with the system through an app or interface.
    - The service receives sensor data and sends commands to actuators.
    - The virtual entities represent the gate and barrier in digital form.
    - The physical entities are monitored by sensors and controlled by actuators.
    - The Arduino device processes data and manages actions.
    - Resources are made available for user interaction and system control.

Step 4. INFORMATION MODEL SPECIFICATION

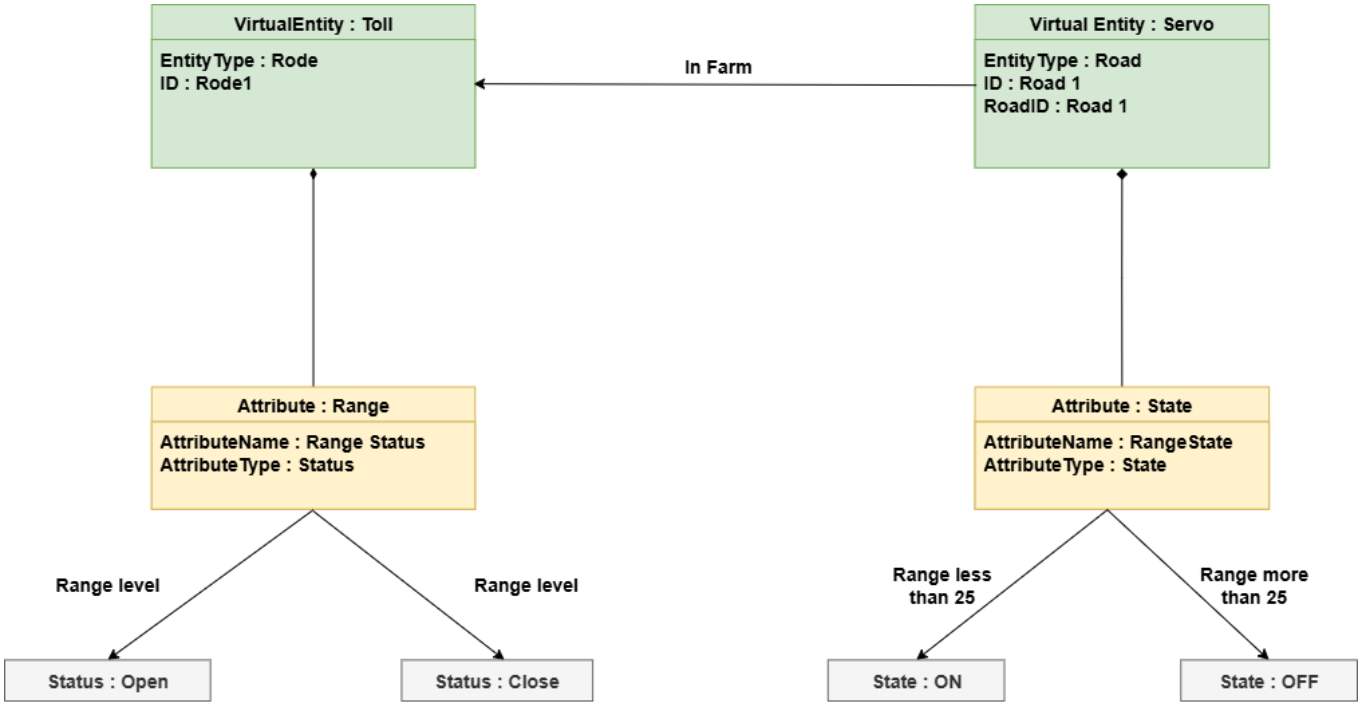
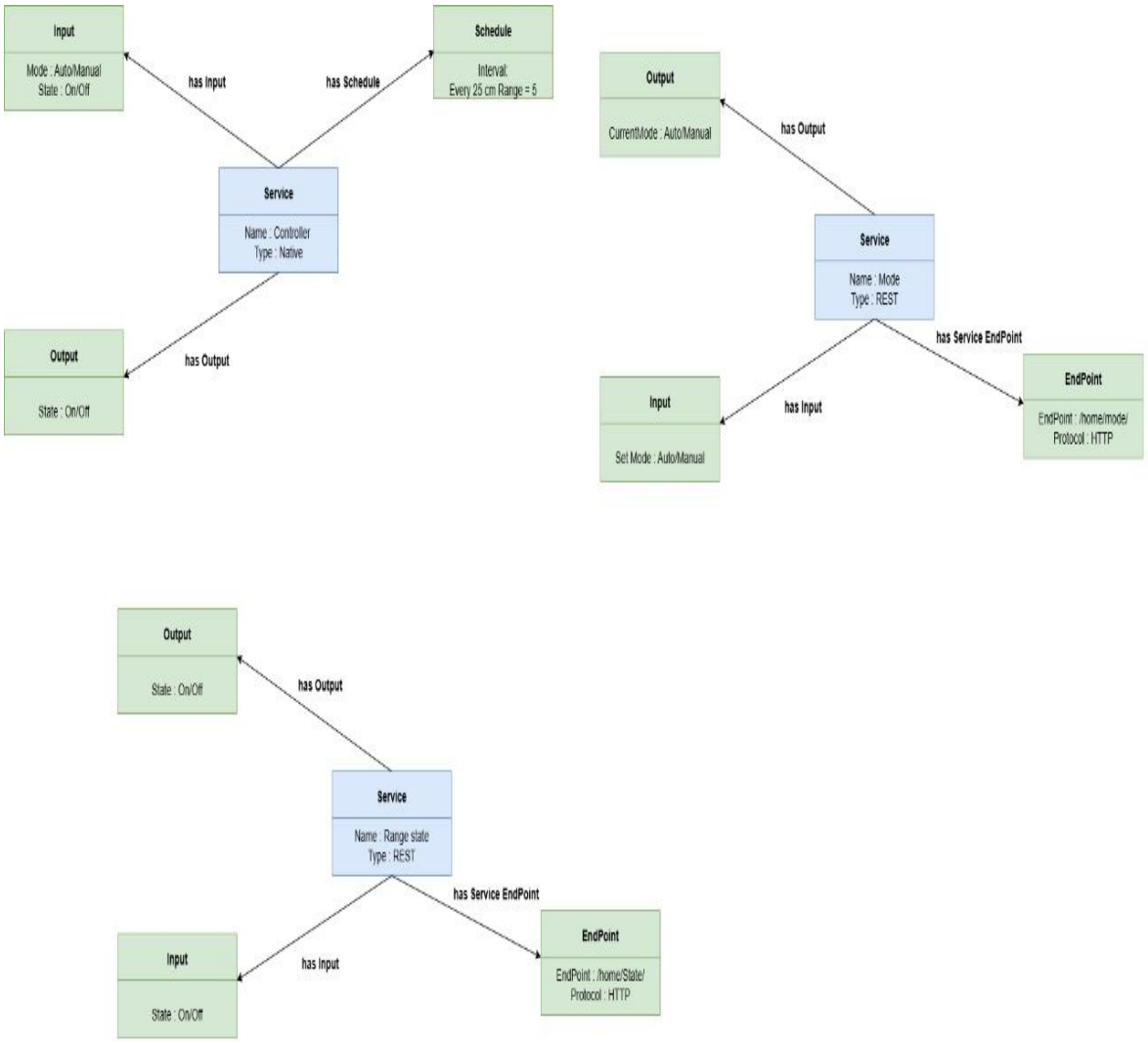


Fig. 4

Step 5. <u>SERVICE SPECIFICATIONS</u>



Step 6. IOT LEVEL SPECIFICATION

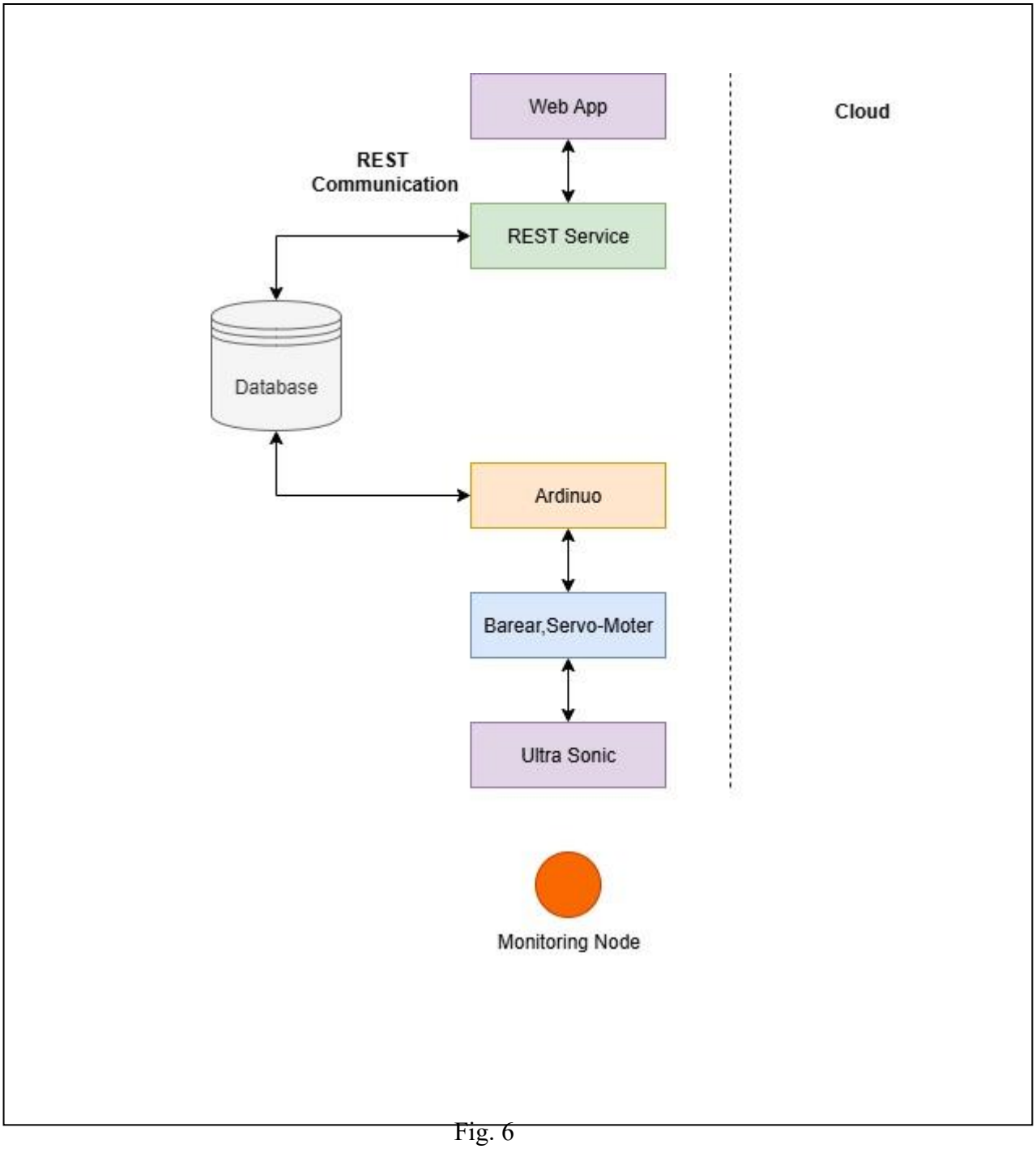


Fig. 6

Step 7. FUNCTIONAL VIEW SPECIFICATION

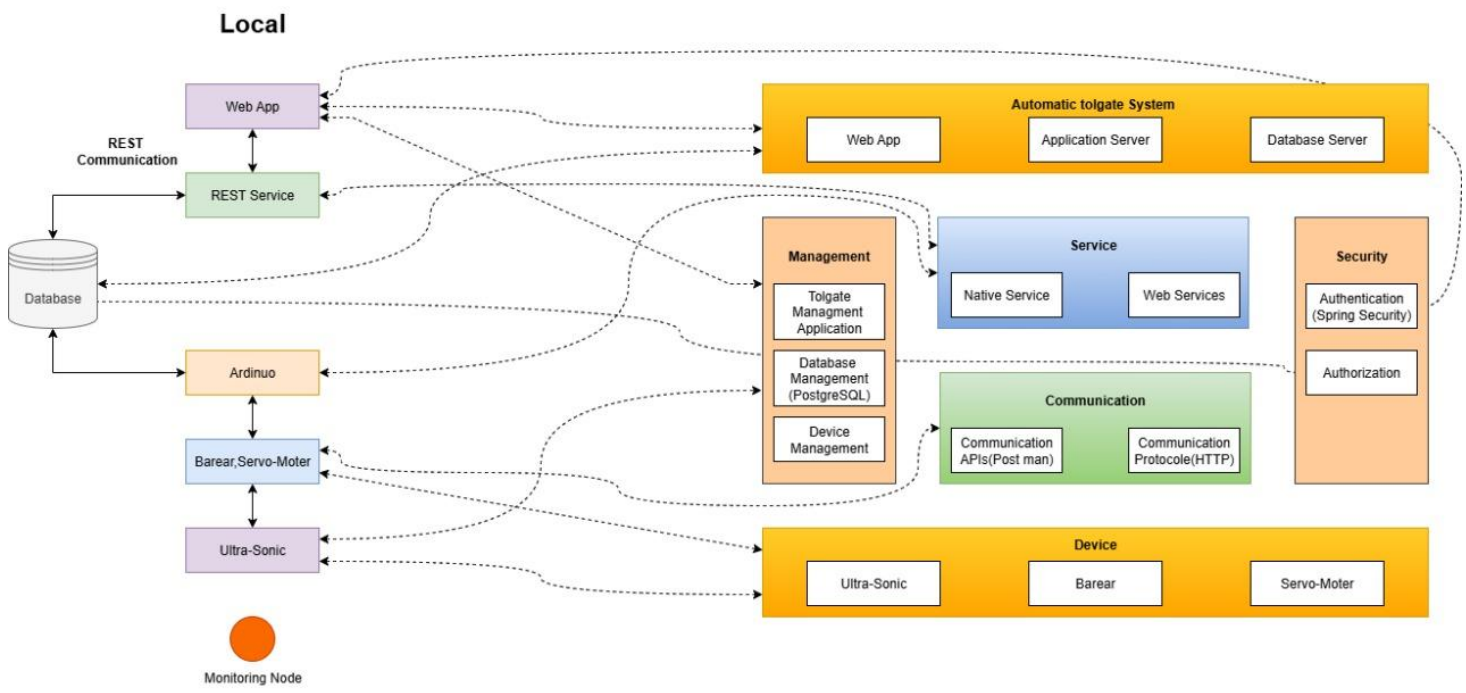


Fig. 7

Step 8. OPERATIONAL VIEW SPECIFICATION

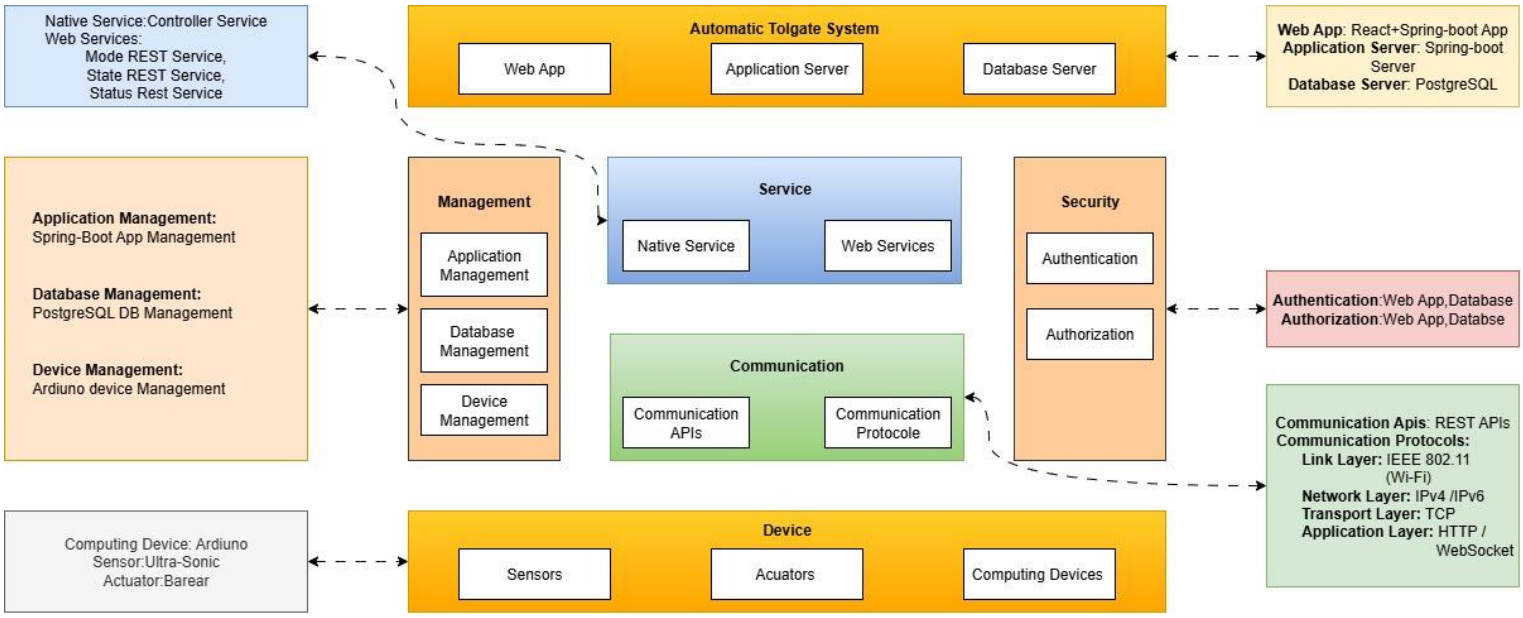


Fig. 8